

The slide features a light blue background. In the top right corner, there is a cluster of dark blue and medium blue triangles. In the bottom right corner, there is a cluster of red and pink triangles. A solid dark blue horizontal bar runs across the bottom of the slide.

Lecture 6:

Sets

Section 2.1

Section Summary

- Definition of sets
- Describing Sets
 - Roster Method
 - Set-Builder Notation
- Some Important Sets in Mathematics
- Empty Set and Universal Set
- Subsets and Set Equality
- Cardinality of Sets
- Tuples
- Cartesian Product

Sets

- A *set* is an unordered collection of objects.
 - the students in this class
 - the chairs in this room
- The objects in a set are called the *elements*, or *members* of the set. A set is said to *contain* its elements.
- The notation $a \in A$ denotes that a is an element of the set A .
- If a is not a member of A , write $a \notin A$

Describing a Set: Roster Method

- $S = \{a, b, c, d\}$
- Order not important
$$S = \{a, b, c, d\} = \{b, c, a, d\}$$
- Each distinct object is either a member or not; listing more than once does not change the set.
$$S = \{a, b, c, d\} = \{a, b, c, b, c, d\}$$
- Elipses (...) may be used to describe a set without listing all of the members when the pattern is clear.
 - $S = \{a, b, c, d, \dots, z\}$

Roster Method

- Set of all vowels in the English alphabet:

$$V = \{a, e, i, o, u\}$$

- Set of all odd positive integers less than 10:

$$O = \{1, 3, 5, 7, 9\}$$

- Set of all positive integers less than 100:

$$S = \{1, 2, 3, \dots, 99\}$$

- Set of all integers less than 0:

$$S = \{\dots, -3, -2, -1\}$$

Some Important Sets

- $\mathbf{N}$ = *natural numbers* = $\{0,1,2,3,\dots\}$
- $\mathbf{Z}$ = *integers* = $\{\dots,-3,-2,-1,0,1,2,3,\dots\}$
- $\mathbf{Z}^+$ = *positive integers* = $\{1,2,3,\dots\}$
- $\mathbf{R}$ = *set of real numbers*
- $\mathbf{R}^+$ = *set of positive real numbers*
- $\mathbf{C}$ = *set of complex numbers.*
- $\mathbf{Q}$ = *set of rational numbers*

Set-Builder Notation

- Specify the property or properties that all members must satisfy:

$$S = \{x \mid x \text{ is a positive integer less than } 100\}$$

$$O = \{x \mid x \text{ is an odd positive integer less than } 10\}$$

$$O = \{x \in \mathbf{Z}^+ \mid x \text{ is odd and } x < 10\}$$

- A predicate may be used:

$$S = \{x \mid P(x)\}$$

- Example: $S = \{x \mid \text{Prime}(x)\}$

- Positive rational numbers:

$$\mathbf{Q}^+ = \{x \in \mathbf{R} \mid x = p/q, \text{ for some positive integers } p, q\}$$

Interval Notation

- $[a,b] = \{x \mid a \leq x \leq b\}$
- $[a,b) = \{x \mid a \leq x < b\}$
- $(a,b] = \{x \mid a < x \leq b\}$
- $(a,b) = \{x \mid a < x < b\}$

closed interval $[a,b]$

open interval (a,b)

Universal Set and Empty Set

- The *universal set* U is the set containing everything currently under consideration.
 - Sometimes implicit
 - Sometimes explicitly stated.
 - Contents depend on the context.
- The *empty set* is the set with no elements.
 - Symbolized as $\emptyset$ or, $\{\}$.

Venn Diagrams

- The universal set is represented by a rectangle.
- Sets are shown as ovals, circles or other geometric shapes.
- Sometimes points are used to represent the particular elements of the set.

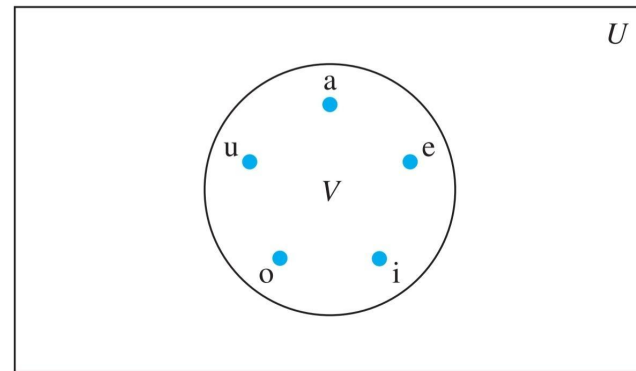
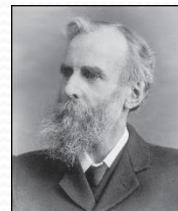


FIGURE 1 Venn diagram for the set of vowels.



John Venn (1834-1923)
Cambridge, UK

Some things to remember

- Sets can be elements of sets.

$\{\{1,2,3\}, a, \{b,c\}\}$

$\{\mathbf{N}, \mathbf{Z}, \mathbf{Q}, \mathbf{R}\}$

- The empty set is different from a set containing the empty set.

$$\emptyset \neq \{ \emptyset \}$$

Set Equality

Definition : Two sets are *equal* if and only if they have the same elements.

- Therefore if A and B are sets, then A and B are equal if and only if $\forall x(x \in A \leftrightarrow x \in B)$
- We write $A = B$ if A and B are equal sets.

$$\{1,3,5\} = \{3, 5, 1\}$$

$$\{1,5,5,5,3,3,1\} = \{1,3,5\}$$

Subsets

Definition : The set A is a *subset* of B , if and only if every element of A is also an element of B .

- The notation $A \subseteq B$ is used to indicate that A is a subset of the set B .
- $A \subseteq B$ holds if and only if $\forall x(x \in A \rightarrow x \in B)$ is true.
 - $\emptyset \subseteq S$, for every set S (Because $a \in \emptyset$ is always false).
 - $S \subseteq S$, for every set S (Because $a \in S \rightarrow a \in S$).

Showing a Set is or is not a Subset of Another Set

- **Showing that A is a Subset of B** : To show that $A \subseteq B$, show that if x belongs to A , then x also belongs to B .
- **Showing that A is not a Subset of B** : To show that A is not a subset of B , $A \not\subseteq B$, find an element $x \in A$ with $x \notin B$. (Such an x is a counterexample to the claim that $x \in A$ implies $x \in B$.)

Examples :

- The set of all computer science majors at your school is a subset of all students at your school.
- The set of integers with squares less than 100 is not a subset of the set of nonnegative integers.

Another look at Equality of Sets

- Recall that two sets A and B are *equal*, denoted by

$$A = B, \text{ iff } \forall x (x \in A \leftrightarrow x \in B)$$

- Using logical equivalences we have that $A = B$ iff

$$\forall x [(x \in A \rightarrow x \in B) \wedge (x \in B \rightarrow x \in A)]$$

- This is equivalent to

$$A \subseteq B \quad \text{and} \quad B \subseteq A$$

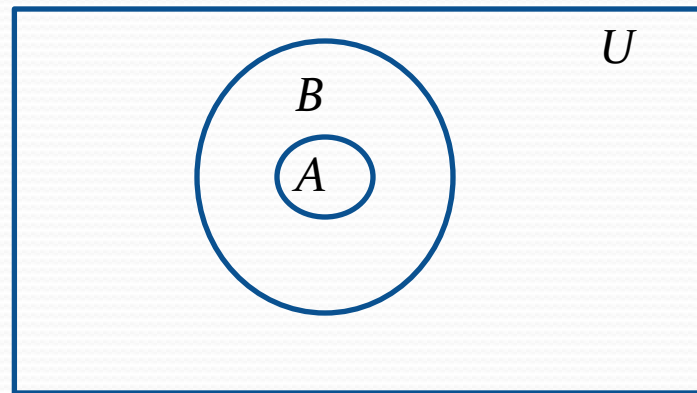
Proper Subsets

Definition : If $A \subseteq B$, but $A \neq B$, then we say A is a *proper subset* of B , denoted by $A \subset B$. If $A \subset B$, then

$$\forall x(x \in A \rightarrow x \in B) \wedge \exists x(x \in B \wedge x \notin A)$$

is true.

Venn Diagram:



Set Cardinality

If there are exactly n distinct elements in S where n is a nonnegative integer, we say that S is *finite*. Otherwise it is *infinite*.

Definition : The *cardinality* of a finite set A , denoted by $|A|$, is the number of (distinct) elements of A .

Examples :

1. $|\emptyset| = 0$
2. Let S be the letters of the English alphabet. Then $|S| = 26$
3. $|\{1,2,3\}| = 3$
4. $|\{\emptyset\}| = 1$
5. The set of integers is infinite.

Power Sets

Definition : The set of all subsets of a set A , denoted $P(A)$, is called the *power set* of A .

Example : If $A = \{a, b\}$ then

$$P(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$$

Example : If $A = \{\emptyset, \{a, b\}\}$ then

$$P(A) = \{\emptyset, \{\emptyset\}, \{\{a, b\}\}, \{\emptyset, \{a, b\}\}\}$$

- If a set has n elements, then the cardinality of the power set is 2^n .



René Descartes
(1596-1650)

Cartesian Product

Definition : The *Cartesian Product* of two sets A and B , denoted by $A \times B$ is the set of ordered pairs (a,b) where $a \in A$ and $b \in B$.

$$A \times B = \{(a, b) | a \in A \wedge b \in B\}$$

Example :

$$A = \{a, b\} \quad B = \{1, 2, 3\}$$

$$A \times B = \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3)\}$$

- **Definition :** A subset R of the Cartesian product $A \times B$ is called a *relation* from the set A to the set B .

Cartesian Product

Definition : The cartesian products of the sets $A_1, A_2, \dots, A_n$ denoted by $A_1 \times A_2 \times \dots \times A_n$, is the set of ordered n -tuples $(a_1, a_2, \dots, a_n)$ where a_i belongs to A_i for $i = 1, \dots, n$.

$$A_1 \times A_2 \times \dots \times A_n = \{(a_1, a_2, \dots, a_n) \mid a_i \in A_i \text{ for } i = 1, 2, \dots, n\}$$

Example : What is $A \times B \times C$ where $A = \{0,1\}$, $B = \{1,2\}$ and $C = \{0,1,2\}$

Solution: $A \times B \times C = \{(0,1,0), (0,1,1), (0,1,2), (0,2,0), (0,2,1), (0,2,2), (1,1,0), (1,1,1), (1,1,2), (1,2,0), (1,2,1), (1,2,2)\}$